Amazon Interview Experience

Amazon is an **American multinational technology** company focusing on ecommerce, cloud computing, online advertising, digital streaming, and artificial intelligence. It is considered one of the **Big Five American technology companies,** alongside Alphabet (parent company of Google), Apple, Meta (parent company of Facebook) and Microsoft. It is a product-based company and one of the infamous **MAANG** companies.

Role Offered: SDE-6M Intern (FTE conversion is based upon performance)

Stipend: 1,10,000 INR + 425 USD / Month

Location : Bangalore / Chennai / Hyderabad (Available Locations)

Selection Process:

1.PrePlacement Talk

2.Online Assessment

3.Technical Interview

Prerequisite:

Don't apply without knowing all concepts from the following

1.Binary Search

2.Graphs

3.Trees

4.Linked List

These topics are mandatory for Amazon Technical interview.
Questions may come from some other topics but, they will ask totally **2** questions in the interview which includes one Linear Data Structure and another one is Non-Linear Data Structure so, definitely you will have one question from the above topics.

Complete all the concepts from the Stiver's A2Z sheet.

Mastering all the concepts from the A2Z sheet will surely boost your chances of cracking Amazon.

Online Assessment (3rd Oct 2025) :(Virtual)

There is no guarantee for shortlist even though you could solve both coding questions.

Personality Test plays a major role for shortlisting. (be minded)

It consisted of two **coding questions** and a work **personality test.**

- Coding Question - 2(A total of 70 minutes)
- Personality test – No time constraint (But compulsory to complete)

The question pattern for the online assessment would change every year. This year it was only coding round whereas previously they would ask coding questions along with some technical questions. So, prepare yourself for both **coding questions** and **MCQ's.**

The online Assessment was conducted in **HackerRank** platform.

No need of **Webcam** and **Microphone** for this OA round.

Coding Question 1:

The logistics coordinator at a **fulfillment center** needs to efficiently dispatch a set of n items, each originating from a specific warehouse, represented by an array of warehouse **ID**s. The objective is to find the **minimum number of operations** required to clear the entire inventory, subject to two rules:

1. **Paired Dispatch:** Two items can be dispatched in a single operation *only if* they are from **different warehouses**.
2. **Single Dispatch:** An item must be dispatched alone if no eligible partner from a different warehouse is available.

The challenge is to devise a strategy that maximizes the use of the more efficient paired dispatch operation to achieve the lowest possible total operation count.

Test Case 1:

Given n = 4 and warehouses = [9,5,9,9]

Operation	X	Y	Warehouses
1.	1	-	[5,9,9]
2.	1	2	[9,9]
3.	1	-	[]

Hence the coordinator needs to perform 3 operations to dispatch all of the items.

Output :3

Explaination:

Optimal Strategy: Majority Rule

The key to minimization lies in focusing on the **most frequently occurring warehouse ID**. Let be the count of items from this majority warehouse, and let be the total count of items from all *other* (minority) warehouses.

You can pair every item from the majority warehouse with an item from the minority group using Operation 1, until one group runs out.

- **If the majority is not too large (M <= S):** You have enough items from other warehouses to pair with every item from the majority warehouse. All M items are paired and dispatched in M operations. The remaining S - M items are all from the minority group and can also be optimally paired with each other. The total minimum operations required ends up being M+[(S-M)/2] .
- **If the majority is dominant (M > S):** You pair all S minority items with items from the majority warehouse, taking S operations. The remaining M- S items are all from the *same* dominant warehouse and must be dispatched one-by-one using Operation 2. The total minimum operations required is S + (M- S) = M .

In essence, the minimum number of operations is dictated by either the necessity of one-by-one dispatching for the **dominant warehouse count (M)** or the optimized two-by-two dispatching possible when item counts are **balanced**.

Coding Question 2:

The task is to determine the time required for a service **pipeline** to stabilize following a disruptive service failure. The pipeline is represented as a string of service categories, and the failure is caused by a pecific **failedService** character. The system evolves in discrete time steps (seconds): in **every time step**, each instance of the **failedService** currently present in the pipeline causes the immediate termination and removal of the service node **immediately preceding** it. This process iteratively shortens and changes the pipeline string. The objective is to calculate the total number of time steps (seconds) that pass until a full step occurs with **no services being terminated**, at which point the system is considered stabilized.

Sample Testcase:

N = 4 , pipeline = "database" , failedService = "a"

Output: 2

Explanation:

Problem Setup :

We have a pipeline represented by the word: **database**

The failed service is: **a**

Rule:
Whenever an 'a' appears in the pipeline, the **letter immediately before it** is terminated (removed).
This process repeats in time steps until no more letters can be removed (the system becomes stable).

Step-by-Step Execution :

Time Step 0 — Initial State :

Position	0	1	2	3	4	5	6	7
Letter	d	a	t	a	b	a	s	e

failedService = a

Occurrences of 'a' are at positions **1, 3, and 5**.

Each ‘a’ removes the letter before it:

- ‘a’ at pos 1 removes **d (pos 0)**
- ‘a’ at pos 3 removes **t (pos 2)**
- ‘a’ at pos 5 removes **b (pos 4)**

Time Step 1 — After First Removals :

Remaining letters (after removing d, t, b):
aaase

Position	0	1	2	3	4
Letter	a	a	a	s	e

Again check for ‘a’:

- ‘a’s are at positions 0, 1, 2.
- There is **no letter before position 0**, so nothing can be removed.

No changes occur in this step.

Time Step 2 — System Stable :

Since no further removals happen, the system has stabilized.

Final Output :

Time Step	Pipeline State
0	database
1	aaase
2	aaase (stable)

Final Stable Pipeline: “**aaase**” .

Number of Time Steps to Stabilize**: 2**

Based on this Online Assessment, 105 people were shortlisted for the next round.

Technical Interview: (1 hour) 10th Oct between 10 AM to 6 PM.

The interview was conducted **in Amazon's chime** platform. We were provided an amazon chime link for the meeting.

In the interview, we had an introduction where each of us took a turn introducing ourselves. In this interview, you wouldn't have any HR round to talk about your strengths. So, when you introduce yourself, tell them your technical strengths and what projects you have worked on. Present yourself in a pleasing way. Then you would be asked to share your screen and use **LiveCode** to code your questions.

1.Land & Sea Problem

You are given a grid consisting only of 0s and 1s, where 0 represents sea and 1 represents land. Some land cells are connected to one another horizontally or vertically, forming connected land regions. If any connected region of land touches the boundary of the grid—meaning the top row, bottom row, leftmost column, or rightmost column—then that entire region becomes flooded. Flooding means that every 1 in that region is converted into a 0. After all boundary-touching land regions are removed in this way, you must determine how many land cells (1s) remain in the grid.

Grid = [

[0,1,0,0,0],
[1,1,1,0,0],
[0,0,1,0,1],
[0,0,0,0,0],
[0,1,1,1,0],
[0,0,0,0,0]
]
Output : 3

I was asked the approach first and then asked to code the same.

Then they enquired me about the time and space complexities.

Explanation:

This problem can be efficiently solved using Depth-First Search (DFS). The key idea is that only the land regions that touch the boundary should be removed, so instead of searching the entire grid, we only start DFS from the boundary cells. We go through all cells on the outer border of the grid, and whenever we find a land cell (a **1**), we immediately apply DFS on it. During the DFS process, we convert that **1** into a **0** to mark it as flooded, and we continue exploring all four directions—up, down, left, and right—so long as the next cell is also land. This ensures that every land cell connected to the boundary is removed. After performing this boundary DFS on all sides, whatever **1**s remain inside the grid are guaranteed to be land regions that never touched the boundary. Therefore, the final answer is simply the count of all remaining **1**s in the grid after the flooding process is complete.

2.Koko Eating Bananas

This problem is available on leetcode :

https://leetcode.com/problems/koko-eating-bananas/description/

I was asked the approach first and then asked to code the same. Then they enquired me about the time complexitiy.

With that the interview got over.

The result was announced on **13|10|2025** and I was one of the **18** members who got the offer.

""" *Everything will happen at the right time* """

" **Just keep working hard and don't lose hope.**
And always have good friends around you —
the ones who motivate you, support you, and pray for you ❤️ 🫂 ... "

Tips:

- Regular practice is important. Solve questions in **Leetcode, GeeksForGeeks** and **HackerRank.**
- I followed **C.O.D.E. resources** and **takeUForward YouTube channel** for my preparation on topics like Trees, Graphs, DP and Recursion.
- If you are not confident with **DBMS, OS** and **OOPS**, then prepare for them simultaneously because you won't be having much time between the online test and the interview.
- Aptitude is as important as DSA. At least start learning aptitude two weeks before your online test.
- Even though I was not asked about my projects, in every interview they will ask about your projects, and they might ask you about the database structure and all. So be prepared to answer any questions related to your projects.
- For any DSA question give them the brute force approach and don't jump to the optimized answer directly. The interviewers are mostly interested in your thought process rather than your code.

In the end, after all the preparation go through **the 14 leadership requirements from Amazon** (just in case, interviewer may check) and **STAR** approach.

All the very best!!!

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